Topic 7: Design (part 3)

SCSM2113 Human-Computer Interaction Fundamentals
Faculty of Computing
Universiti Teknologi Malaysia

1

01 UI ELEMENTS IN GENERAL

02 ERROR HANDLING & PREVENTION

03 PREVENTING BOTH MISTAKES & SLIPS

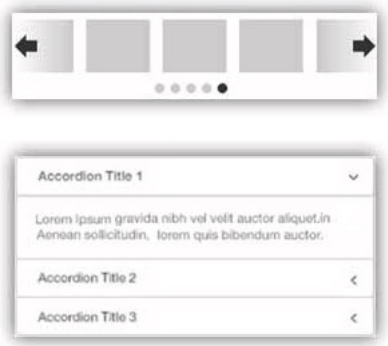
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UI ELEMENTS IN GENERAL

3

UI Elements

- Interface elements include but are not limited to:
  - Input Controls: checkboxes, radio buttons, dropdown lists, list boxes, buttons, toggles, text fields, date field
  - Navigational Components: breadcrumb, slider, search field, pagination, slider, tags, icons, image carousel
  - Informational Components: tooltips, icons, progress bar, notifications, message boxes, modal windows
  - Containers: accordion



4

ERROR HANDLING & PREVENTION

SLIPS

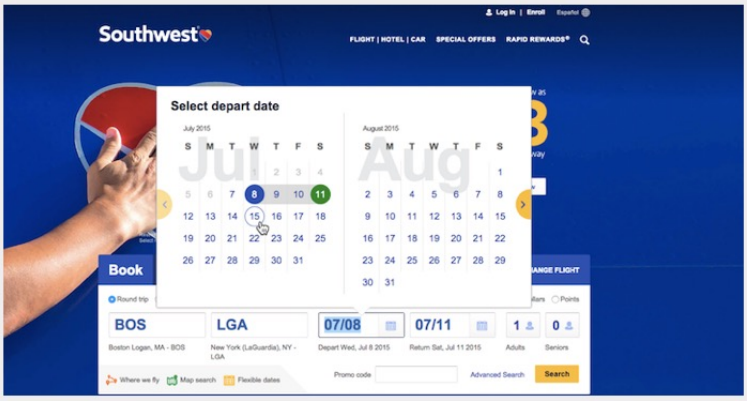
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Slips

- Slips occur when users intend to perform one action, but end up doing another (often similar) action.
- For example, typing an “i” instead of an “o” counts as a slip; accidentally putting liquid hand soap on one’s toothbrush instead of toothpaste is also a slip.
- Slips are typically made when users are on autopilot, and when they do not fully devote their attention resources to the task at hand

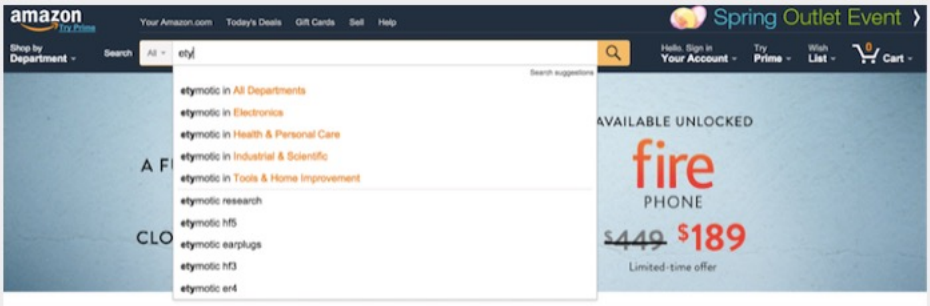
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Include Helpful Constraints



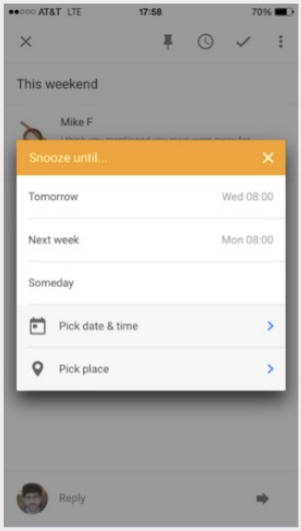
Southwest's calendar widget for picking flight dates uses helpful constraints to prevent users from accidentally setting a nonsensical date range. Even if users attempt to set the return date before the departure date, this widget forces them to pick a departure date first. In addition, it subtly uses color to provide context about which date is about to be changed (in this case, blue for departure), which helps users see which field they are selecting (instead of having to keep that information in their working memory).

Offer Suggestions



Remembering how to spell Etymotic Research *is difficult for users searching for high-quality headphones, and typing is likely to be low accuracy as well. Amazon's clickable search suggestions enable users to type less, thereby making fewer slips or mistakes that would produce no results.*

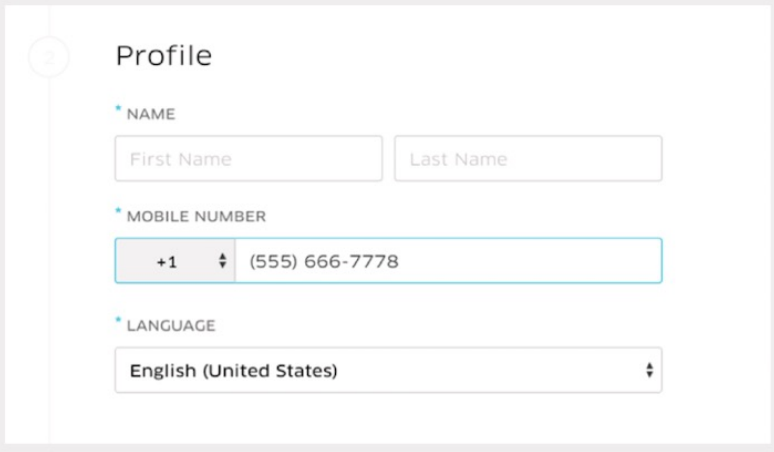
Choose Good Defaults



Google's Inbox app for iOS allows you to "snooze" an email until a later time. The default options are sensible and prevent typing errors for common choices.

9

Use Forgiving Formatting



Uber.com automatically displays the phone number in the desired format as users type, so that they can more easily scan their work to confirm that it's correct.

10

10

ERROR HANDLING & PREVENTION
MISTAKES

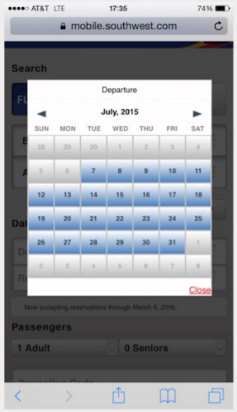
11

Mistakes

- Mistakes are made when users have goals that are inappropriate for the current problem or task; even if they take the right steps to complete their goals, the steps will result in an error.
- For example, if I misunderstood the meaning of the oil-pressure warning light in my car, and thought it was the tire-pressure monitor, no matter how carefully I added air to my tires, it would not fix the issue with my oil pressure.
- This would be a mistake, since the goal that I was attempting to accomplish was inappropriate for the situation, even though I made no errors in executing my plan.
- Mistakes are conscious errors, and often (though not exclusively) arise when a user has incomplete or incorrect information about the task, and develops a mental model that doesn't match how the interface actually works.

12

Gather User Data (follow Design Conventions)



Southwest's mobile site uses the convention of greying out dates in the past, letting you know you can't select them when booking a flight. Unfortunately, it also uses the same design for the next month's dates, implying unavailability.

13

13

Communicate Affordance

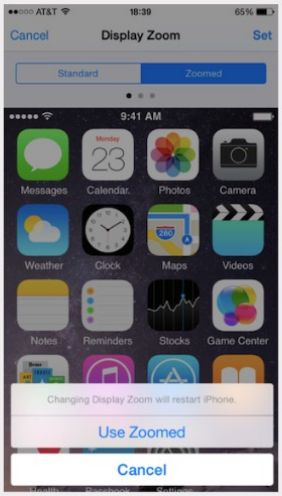


Polarr is a popular photo editor on iOS. The right-hand editing controls (Temp, Tint, etc.) require that you tap on the box, and then swipe left or right to change the numeric value of that parameter. However, the controls don't indicate how users should interact with them, so novices are likely to accidentally set those controls to the wrong values several times before learning how to interact with them properly. An additional interaction difficulty is present: since the controls are on the far right side of the display, you can easily decrease the parameters by swiping left, but you quickly run out of horizontal space to swipe right and increase the value.

14

14

Preview Results



*In iOS 8, there is an accessibility option that allows users with low vision to zoom the display so that icons and text are larger. Applying zoom requires restarting the phone, which is a **heavyweight** action that will take a while, so iOS shows a preview of what things will look like before you commit to applying this change. This helpful preview allows you to evaluate whether or not your goal really is to zoom the display.*

15

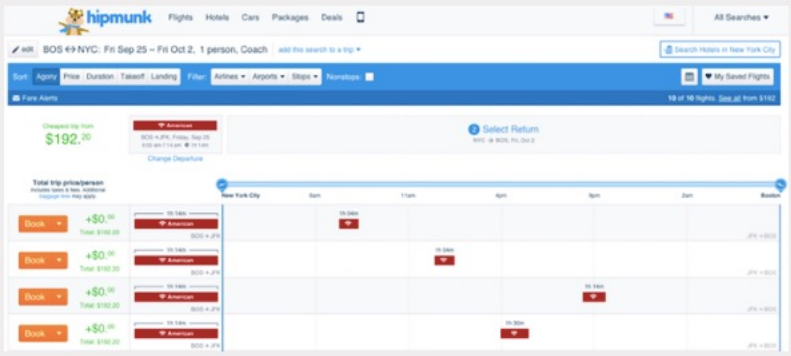
ERROR HANDLING & PREVENTION

MISTAKES

16

16

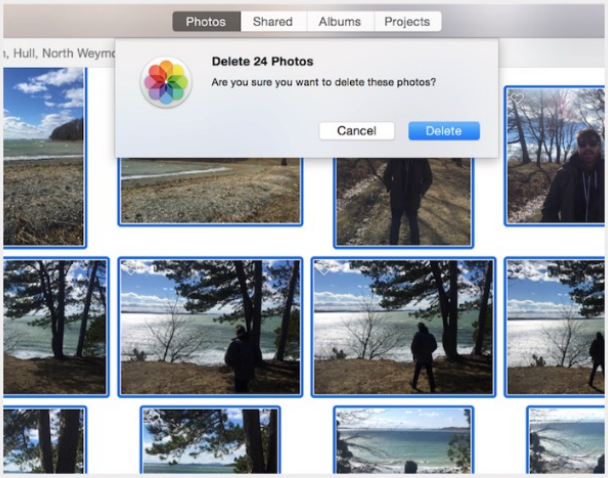
Remove Memory Burdens



Hipmunk provides a quick glance at the contextual information needed to resume the process of choosing a flight, even after an interruption. At this second step in the booking process, it clearly shows the dates of travel, the airports in question, the fact that the lowest-priced fare for departure was chosen, and that the user is required to pick a return flight. Even users who had been distracted for quite some time could easily resume this process without accidentally deviating from their original requirements and plan for this flight, or attempting to repeat the steps already completed.

17

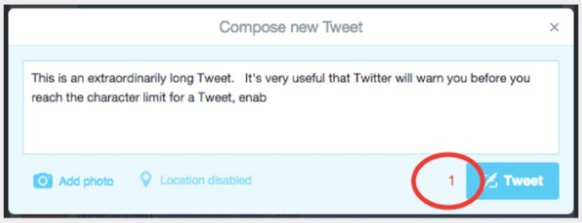
Confirm before Destructive Actions



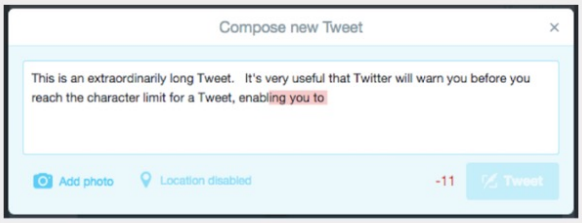
Apple's new Photos app uses a conventional dialog box to make sure that the user really intended to delete these photos of a recent hiking trip, and indicates that the scope of the action is the highlighted 24 photos. Even better, the button to confirm the delete action is clearly labeled Delete, *rather than the generic* Confirm.

18

Warn before Errors are Made



Twitter famously has a strict character limit for Tweets, and warns users before they exceed that limit with a remaining character count.



Once a Tweet is longer than the limit, it shows a negative counter, highlights the excess characters, and deactivates the Tweet button, which lets users know exactly what they need to do to fix their mistake.

19